

CIRCULATION IN LERAY VARIABLES: GATHERING, LOCAL ENERGY, AND AN OUTGOING TRANSPORT LAW

ADILKHAN SALKIMBAYEV

ABSTRACT. In backward similarity variables the vorticity of a Navier–Stokes solution is frozen into the outward field $V = U + \xi/2$. We record what this, together with the leak law for strained vortices, says about a vortex tube approaching a singular point. (1) If the tube’s accumulated strain time τ diverges, the cumulative leak is infinite unless $\Gamma_c/\nu \gtrsim \frac{2\pi\lambda}{c(\lambda)} \ln \tau$ ($\approx 54 \ln \tau$ at $\lambda = 2$): a tube that survives must *gather* circulation. (2) An elementary inequality bounds averaged circulation by the scale-invariant local energy, $\langle |\Gamma| \rangle \leq 3.65\sqrt{A}$, so gathering requires $A \gtrsim (\ln \tau)^2$. For a model gathering tube this places the scenario in Seregin’s weighted class with a slowly varying weight, and his condition (2.4) holds for every admissible exponent, so his Theorem 2.1 excludes it whenever his weighted energy bound holds with the natural weight. (3) Under weak- L^3 control outside the backward paraboloid (the hypothesis of Barker–Fernández-Dalgo–Prange), circulation is bounded and gathering is impossible. (4) Under the same control, sphere-averaged circulation obeys $\partial_\tau \vec{\gamma} + \frac{1}{2} \partial_{\ln R} \vec{\gamma} = O(R^{-3/2})$: outward transport, with no reservoir beyond a finite Leray radius. Known necessary conditions suggest the control in (3)–(4) fails at actual singularities; we say what this leaves.

1. INTRODUCTION

Exactly self-similar blowup of the Navier–Stokes equations is excluded (Nečas–Růžička–Šverák [4]; Tsai [9]); discretely self-similar blowup is excluded for scaling factors near 1 (Chae–Wolf [3]) and, under Type I bounds, for extreme rotation or factor near 1 (Pineau–Vicol [6]). Type II scenarios with slowly growing scale-invariant quantities are the subject of Seregin’s recent work [7, 8], and Barker, Fernández-Dalgo and Prange [1] show that critical norms outside the backward paraboloid must blow up.

This paper looks at the same problem through the circulation of a vortex tube. Two inputs come from companion papers. A tube in a strain whose in-plane part exceeds its axial stretching ($\lambda > 1$, the tearing regime) loses a fraction $\kappa(\varepsilon) \geq e^{-c(\lambda)/\varepsilon}$ of its circulation per unit of strain time, with $\varepsilon = 2\pi\lambda\nu/\Gamma_c$ and a certified $c(2) \leq 0.2331$ [11]; and its centroid drifts off the stagnation point at a rate independent of its structure [12]. Section 2 sets up Leray variables. Section 3 turns the leak into a rate-free criterion: survival needs logarithmic gathering. Section 4 bounds gathered circulation by local energy. Section 5 places gathering in Seregin’s framework, Section 6 closes it under the hypothesis of [1], and Section 7 proves an outgoing transport law under the same hypothesis. Section 9 says why the hypothesis probably fails at real singularities and what remains.

2. LERAY VARIABLES AND THE FLUX LAW

Let u solve the Navier–Stokes equations with $\nu = 1$ and a singular point at $(0, T_*)$. Put $\xi = x/\sqrt{T_* - t}$, $\tau = -\ln(T_* - t)$, $u = (T_* - t)^{-1/2}U$, $\omega = (T_* - t)^{-1}\Omega$.

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Lemma 2.1 (frozen-in outward field). $\Omega_\tau = \nabla \times (V \times \Omega) + \Delta\Omega$ with $V = U + \frac{\xi}{2}$. Fluid particles move by $\frac{d\xi}{d\tau} = V$, and $\nabla \cdot V = \frac{3}{2}$. For any fixed surface D with boundary,

$$\frac{d}{d\tau} \int_D \Omega \cdot n \, dS = \oint_{\partial D} (V \times \Omega - \nabla \times \Omega) \cdot dl.$$

Proof. Differentiating, $\partial_t \omega = (T_* - t)^{-2}(\Omega_\tau + \Omega + \frac{1}{2}\xi \cdot \nabla \Omega)$, while $\nabla_x \times (u \times \omega) = (T_* - t)^{-2} \nabla \times (U \times \Omega)$ and $\Delta_x \omega = (T_* - t)^{-2} \Delta \Omega$. So $\Omega_\tau = \nabla \times (U \times \Omega) + \Delta \Omega - \Omega - \frac{1}{2}\xi \cdot \nabla \Omega$, and $\nabla \times (\frac{\xi}{2} \times \Omega) = \frac{1}{2}(\xi \nabla \cdot \Omega - \Omega \nabla \cdot \xi + (\Omega \cdot \nabla)\xi - (\xi \cdot \nabla)\Omega) = -\Omega - \frac{1}{2}\xi \cdot \nabla \Omega$. The flux law follows from $\Delta \Omega = -\nabla \times \nabla \times \Omega$ and Stokes' theorem. $\square$

At large $|\xi|$ the term $\xi/2$ dominates any velocity that decays: the shrinking parabolic region outruns the fluid, and vorticity is carried outward.

3. THE RATE-FREE LEAK CRITERION

Consider a core tube aligned with an eigenvector of the external strain, with axial strain $s(t) > 0$, in-plane parameter $\lambda \in (1, 3]$, and circulation $\Gamma_c(t)$. Let $\tau_c(t) = \int^t s \, dt'$ be its accumulated strain time. Under a Type I bound $s = \alpha/(T_* - t)$ with $\langle \alpha \rangle > 0$, $\tau_c = \langle \alpha \rangle \ln \frac{1}{T_* - t} + O(1)$. In general, each halving of the core radius costs a strain time of order one, so $\tau_c \rightarrow \infty$ at any singularity at which the core shrinks to zero [10].

Proposition 3.1 (rate-free leak criterion). *Suppose the tube is coherent in the sense of hypothesis (H) of [11] along its history, so that it loses at least the fraction $\kappa(\varepsilon) \, d\tau_c$ per strain time with $\kappa(\varepsilon) \geq e^{-(c+o(1))/\varepsilon}$, $c = c(\lambda) \leq 0.2331$ at $\lambda = 2$. Write $\Gamma_c/\nu = \frac{2\pi\lambda}{c}(\ln \tau_c + h(\tau_c))$. Then the cumulative fraction lost, $\int \kappa \, d\tau_c$, diverges whenever $\int^\infty e^{-h(\tau)} \frac{d\tau}{\tau} = \infty$; in particular whenever $\Gamma_c/\nu \leq \frac{2\pi\lambda}{c} \ln \tau_c$ eventually. At $\lambda = 2$, $\frac{2\pi\lambda}{c} \geq \frac{4\pi}{0.2331} = 53.9$.*

Proof. $\kappa \geq \exp(-c\Gamma_c/2\pi\lambda\nu) = \tau_c^{-1} e^{-h(\tau_c)}$ up to the $o(1)$ in the exponent, which is absorbed in h . $\square$

A core that is not resupplied therefore dies, and a core that survives on its own circulation must *gather*: $\Gamma_c/\nu \gtrsim 54 \ln \tau_c$, i.e. $\gtrsim 54 \ln \ln \frac{1}{T_* - t}$ under Type I scaling. The criterion uses only τ_c ; Type I versus Type II changes only how τ_c relates to t . Gathering is invisible to the leak but not to the drift, which is independent of Γ_c [12].

4. CIRCULATION AND LOCAL ENERGY

Let $A(r) = r^{-1} \int_{B_r} |u|^2 \, dx$, the scale-invariant local energy of the partial-regularity and Type I theory.

Lemma 4.1 (circulation is bounded by local energy). *Let $\theta > 0$, $\theta' = \theta/(2\sqrt{2})$, and fix an axis through the centre of B_θ . For $\rho \in [\theta', 2\theta']$ and z in an axial interval of length $2\theta'$ centred at the centre, let $\Gamma(\rho, z)$ be the circulation of u around the circle of radius ρ normal to the axis at height z . Then*

$$\langle |\Gamma| \rangle := \frac{1}{2\theta'} \int \frac{1}{\theta'} \int_{\theta'}^{2\theta'} |\Gamma(\rho, z)| \, d\rho \, dz \leq \sqrt{3\pi\sqrt{2}} \sqrt{A(\theta)} \approx 3.65 \sqrt{A(\theta)}.$$

Proof. $|\Gamma(\rho, z)| \leq \int_0^{2\pi} |u| \rho \, d\phi$, so the ρ -average is at most $\frac{1}{\theta'} \int_{\text{annulus}} |u| \, dA \leq \frac{1}{\theta'} (3\pi\theta'^2)^{1/2} (\int_{\text{annulus}} |u|^2 \, dA)^{1/2}$. Averaging in z and using Cauchy–Schwarz again, $\langle |\Gamma| \rangle \leq (3\pi)^{1/2} (2\theta')^{-1/2} (\int_{\text{cyl}} |u|^2)^{1/2}$, where the cylinder (radius $2\theta'$, height $2\theta'$) lies in $B(\sqrt{5}\theta') \subset B_\theta$. Hence $\langle |\Gamma| \rangle \leq (3\pi)^{1/2} (2\theta')^{-1/2} (\theta A(\theta))^{1/2} = \sqrt{3\pi\sqrt{2}} \sqrt{A(\theta)}$. $\square$

Corollary 4.2. *If a tube of circulation Γ_c threads all these discs and no vorticity of the opposite sign comparable to Γ_c lies within $2\theta'$ of its axis, so that $\Gamma(\rho, z) \geq \Gamma_c/2$ on them, then $\Gamma_c \leq 7.3\sqrt{A(\theta)}$. Consequently, a coherent tearing-regime tube that survives on its own circulation (Proposition 3.1) requires*

$$A(\theta) \gtrsim (7.4\nu \ln \tau_c)^2 \quad (\lambda = 2)$$

at every scale θ above the core: the local energy ratio must grow at least like $(\ln \tau_c)^2$.

Energy alone allows $A(r) \leq E_0/r$; the gap between this power law and $(\ln \tau)^2$ is the supercriticality gap in this currency. Any bound on the growth of A at a singular point beating $(\ln \tau)^2$ excludes gathering of isolated tubes.

5. SEREGIN'S WEIGHTED FRAMEWORK

Seregin [8] studies Type II scenarios defined by two conditions: a weighted energy bound

$$\sup_{0 < r < 1} \{A_f(v, r) + E_f(v, r) + D_f(q, r)\} < \infty \quad (1.7), \quad A_f = \sup_t \frac{f(r)^2}{r} \int_{B(r)} |v|^2,$$

with $f \downarrow 0$, E_f and D_f the analogous dissipation and pressure quantities; and an intensity condition $g(r_k)M_{\kappa}^{s,l}(v, r_k) \geq \varepsilon_0$ along $r_k \rightarrow 0$ with $g \rightarrow 0$, where $M_{\kappa}^{s,l}(v, a) = a^{-\kappa} \int_{-a^2}^0 (\int_{B(a)} |v|^s)^{l/s} dt$, $\kappa = l(\frac{3}{s} + \frac{2}{l} - 1)$, and $l > \kappa > 0$. His Theorem 2.1: if (1.7) holds, $s < p(\eta)$, $l < q(\eta)$ with $\frac{1}{p(\eta)} = \frac{\eta}{6} + \frac{3(1-\eta)}{10}$, $\frac{1}{q(\eta)} = \frac{\eta}{2} + \frac{3(1-\eta)}{10}$, $0 \leq \eta \leq 1$, the weight satisfies $F(a) = \liminf_{\lambda \rightarrow 0} f(\lambda a)/f(\lambda) > 0$ (2.2), and

$$f(\lambda)^{\frac{l}{2}(1+\frac{3}{s})-\frac{3}{2}(1-\frac{s}{p(\eta)})\frac{l}{s}} / g(\lambda\sqrt{f(\lambda)}) \rightarrow \infty \quad (2.4),$$

then the scenario is impossible. His Theorem 3.1 reduces the complementary case, by an Euler rescaling, to a nontrivial ancient Euler solution on $\mathbb{R}^3 \times (-\infty, 0)$ with F -weighted scale-invariant bounds.

The model gathering tube. Take velocity $|v| \sim L(r)/r$ at scale r , with $L(r) = \Gamma_c/\nu \rightarrow \infty$ slowly varying (at the minimal rate of Proposition 3.1, $L \approx 54 \ln \ln \frac{1}{r}$). Then $A(r) \sim L^2$, so (1.7) can hold only with $f \lesssim 1/L$, and $M_{\kappa}^{s,l}(v, r) \sim L^l$ (the powers of r cancel by the definition of κ), so the intensity condition holds with $g = \varepsilon_0 L^{-l}$.

Proposition 5.1. *For the model gathering tube, if (1.7) holds with $f = c/L$, then (2.2) holds with $F \equiv 1$, and (2.4) holds for every admissible (s, l, η) . Hence Seregin's Theorem 2.1 excludes the scenario.*

Proof. L is slowly varying, so $f(\lambda a)/f(\lambda) \rightarrow 1$ and $g(\lambda\sqrt{f(\lambda)}) \sim g(\lambda)$. The exponent in (2.4) simplifies: $\frac{l}{2}(1+\frac{3}{s}) - \frac{3}{2}(\frac{1}{s} - \frac{1}{p})l = \frac{l}{2}(1+\frac{3}{p})$, independent of s . So the left side of (2.4) is $\sim L^{l-\frac{l}{2}(1+3/p)}$, which tends to infinity iff $p > 3$, and $p(\eta) \in [\frac{10}{3}, 6]$. Admissible exponents exist, e.g. $\eta = 0$, $s = l = 3.2$: $\frac{3}{s} + \frac{2}{l} = 1.5625 \in (1, 2)$. $\square$

Remark 5.2. What escapes Proposition 5.1: gathering scenarios in which (1.7) needs a faster-decaying weight than $1/L$, because the dissipation or the pressure quantity grows faster than the kinetic one, or in which the intensity is not carried by an isolated tube. For those, Theorem 3.1 of [8] applies under its condition (3.3), and for slowly varying f the limit is an ancient Euler solution with *ordinary* bounded scale-invariant energy, dissipation and pressure quantities ($F \equiv 1$): the Euler analogue of the Liouville problem of Koch–Nadirashvili–Seregin–Šverák, which is open. In that limit the leak is gone (viscosity is scaled out), but the exact drift of [12] is an inviscid Biot–Savart fact and survives.

6. THE BRIDGE WITH BARKER–FERNÁNDEZ-DALGO–PRANGE

Let $(0, 0)$ be a first-time singularity of a finite-energy solution, $t < 0$, and $\theta_a(t) = \sqrt{a(-t)}$ with $a = \lambda_S(B_1) > \pi$ as in [1].

Hypothesis 6.1 (H_{out}). $\sup_t \|u(\cdot, t)\|_{L^{3,\infty}(\theta_a(t) < |x| < \sqrt{a})} \leq M$.

Since $1/|x| \in L^{3,\infty}$, (H_{out}) is exactly Type I-like decay $|U| \lesssim 1/|\xi|$ of the Leray profile at intermediate scales, in the weak sense.

Theorem 6.2. *Under (H_{out}):*

- (1) [1, Prop. 3.1] $A(\theta_a(t)) \leq A_M$ uniformly in t ;
- (2) every tube satisfying the isolation condition of Corollary 4.2 at scale $\theta_a(t)$ has $\Gamma_c \leq 7.3\sqrt{A_M}$; gathering is impossible;
- (3) every coherent tearing-regime tube near the singularity has $\varepsilon \geq \varepsilon_* = 2\pi\lambda\nu/(7.3\sqrt{A_M}) > 0$ and loses at least the fraction $\kappa(\varepsilon_*) > 0$ per unit strain time, at every scale: an infinite cumulative tax.

Proof. (1) is [1, Prop. 3.1]; (2) is Corollary 4.2 with $\theta = \theta_a(t)$; (3) is (2) with the one-sided leak law of [11]. $\square$

Remark 6.3 (isolation). If vorticity of the opposite sign comparable to Γ_c lies within $\theta/(2\sqrt{2})$ of the core, the strain-driving neighbourhood is a mixed-sign cluster, a different branch of the analysis (the companion covering paper [13]). Opposite-sign neighbours are also the only channel by which the drift of a tearing-regime core can be suppressed for long [12]; they supply strain but not circulation.

7. OUTGOING TRANSPORT

Theorem 7.1. *Assume (H_{out}). Let $\psi_R(\xi) = \psi_0(|\xi|/R)$ with ψ_0 smooth, $\psi_0 = 1$ on $[0, 1]$, $\psi_0 = 0$ on $[2, \infty)$, and $\vec{\gamma}(R, \tau) = R^{-1} \int \psi_R \Omega d\xi$. Then*

$$\partial_\tau \vec{\gamma} + \frac{1}{2} \partial_{\ln R} \vec{\gamma} = \vec{E}, \quad \sup_{\tau} \int_{\tau}^{\tau+1} |\vec{E}| d\tau' \leq C(M) R^{-3/2} \quad \text{for } 2R \leq N\sqrt{a},$$

with $N = N(M)$ from [1], at late τ .

Proof. Let $G_R = \int \psi_R \Omega$. By Lemma 2.1, $\frac{d}{d\tau} G_R = I_1 + I_2 + I_3$ with $I_1 = \int \psi_R \nabla \times (\frac{\xi}{2} \times \Omega)$, $I_2 = \int \psi_R \nabla \times (U \times \Omega)$, $I_3 = \int \psi_R \Delta \Omega$.

I_1 (exact). By the computation in Lemma 2.1, $I_1 = \int \psi_R (-\Omega - \frac{1}{2} \xi \cdot \nabla \Omega) = -G_R + \frac{1}{2} \int \Omega \nabla \cdot (\psi_R \xi) = \frac{1}{2} G_R + \frac{1}{2} \int \Omega \xi \cdot \nabla \psi_R$, and $\xi \cdot \nabla \psi_R = -R \partial_R \psi_R$, so $I_1 = \frac{1}{2} G_R - \frac{R}{2} \partial_R G_R$. For $\vec{\gamma} = G_R/R$ this contributes exactly $-\frac{1}{2} \partial_{\ln R} \vec{\gamma}$ to $\partial_\tau \vec{\gamma}$.

I_2 . Integrating by parts, $|I_2| \leq \int |U| |\Omega| |\nabla \psi_R| \leq \|\psi'_0\|_\infty R^{-1} \int_{S_R} |U| |\Omega|$, $S_R = \{R \leq |\xi| \leq 2R\}$. By Hölder's inequality in Lorentz spaces (O'Neil; [1, Prop. A.1]), $\int_{S_R} |U| |\Omega| \leq C \|U\|_{L^{3,\infty}(S_R)} \|\Omega\|_{L^{3/2,1}(S_R)}$, and on a set of finite measure $\|\Omega\|_{L^{3/2,1}(S_R)} \leq C |S_R|^{1/6} \|\Omega\|_{L^2(S_R)} \leq C R^{1/2} \|\Omega\|_{L^2(S_R)}$. In Leray variables the annulus of (H_{out}) is $\sqrt{a} < |\xi| < \sqrt{a} e^{\tau/2}$, which contains S_R at late times, and weak- L^3 is scale invariant, so $\|U\|_{L^{3,\infty}(S_R)} \leq M$. Hence $|I_2| \leq C M R^{-1/2} \|\Omega\|_{L^2(S_R)}$.

I_3 . $|I_3| = |\int \Omega \Delta \psi_R| \leq \|\Omega\|_{L^2(S_R)} \|\Delta \psi_R\|_{L^2} \leq C R^{-1/2} \|\Omega\|_{L^2(S_R)}$.

Dividing by R , $|\vec{E}| \leq C(1 + M) R^{-3/2} \|\Omega\|_{L^2(S_R)}$. Finally, the weighted energy estimate of [1, §4, Step 1] with weight exponent $\gamma = 0$ reads $\theta_a(s)^{-1} \int_s^0 \|\nabla(\Psi_N u)\|_{L^2}^2 dt \leq C(M)$ for

s close to 0. In Leray variables ($\|\nabla u\|_{L^2(B(N\theta))}^2 = (-t)^{-1/2}\|\nabla U\|_{L^2(B_{N\sqrt{a}})}^2$, $dt = (-t)d\sigma$, $\theta_a(s) = \sqrt{a}e^{-\sigma/2}$) this is $a^{-1/2}\int_{\sigma_s}^{\infty} e^{-(\sigma-\sigma_s)/2}\|\nabla U\|_{L^2(B_{N\sqrt{a}})}^2 d\sigma \leq C(M)$ for every late σ_s , hence $\sup_{\sigma} \int_{\sigma}^{\sigma+1} \|\nabla U\|_{L^2(B_{N\sqrt{a}})}^2 \leq \sqrt{ea}C(M)$. Cauchy–Schwarz in time gives the claim. $\square$

Corollary 7.2 (no reservoir beyond a finite Leray radius). *Under (H_{out}) , the characteristics of $\partial_{\tau} + \frac{1}{2}\partial_{\ln R}$ point strictly outward. For $R \geq R_0$, $\vec{\gamma}(R, \tau)$ is, up to $C(M)R_0^{-3/2}$ per unit τ , the outward-advected record of what crossed R_0 earlier. If a tube through the origin pays the tax of Theorem 6.2(3), at rate $\geq \kappa_*\Gamma_c$ per unit strain time, and $C(M)R_0^{-3/2} < \frac{1}{2}\kappa_*\langle\alpha\rangle\Gamma_c$, then its resupply cannot come from beyond R_0 : it must be recycled inside B_{R_0} , or come with vorticity of the opposite sign.*

Sketch. The transport equation has speed $\frac{1}{2}$ in $\ln R$ and an error integrable at rate $C(M)R^{-3/2}$. Inside B_{R_0} the aligned component of $\vec{\gamma}$ changes only through the vorticity flux across $|\xi| \approx R_0$; with vorticity of the core’s sign there, it can only decrease up to the error. It can increase only when anti-aligned vorticity is swept out. $\square$

Remark 7.3. The spherical weight has no preferred axis, so tilting and stretching by ∇U sit inside I_2 and are bounded the same way. [1] work with first-time singularities of finite-energy solutions and note that their result localises to suitable weak solutions; the first-time setting suffices for the global regularity problem.

8. DISCRETELY SELF-SIMILAR PROFILES

For a backward discretely self-similar solution with factor $\Lambda_* > 1$ the Leray profile is periodic in τ with period $T_p = 2 \ln \Lambda_*$. Then: (i) by Lemma 2.1 the period average of the flux through every fixed surface vanishes; (ii) a coherent tearing-regime tube loses at least the fraction $q = 1 - \exp(-\int_0^{T_p} \kappa \alpha d\tau) > 0$ of its circulation per period, and periodicity forces an equal resupply each period, forever; (iii) its centroid offset grows by $\exp(\frac{\Lambda_*-1}{2}\langle\alpha\rangle T_p)$ per period [12], so the profile has at least one unstable direction per tearing-regime tube unless an exact symmetry pins it; (iv) by Corollary 7.2, under (H_{out}) the resupply is recycled inside a bounded Leray ball. Such a profile is a periodic orbit with a bounded recirculation and finite codimension: non-generic, not excluded. Its exclusion is a Liouville statement for discretely self-similar profiles of the kind proved for factors near 1 in [3, 6].

9. ASSESSMENT

Theorem 6.2 and Theorem 7.1 cover the solutions satisfying (H_{out}) . The known necessary conditions at a singularity point the other way: $\|u(t)\|_{L^{3,q}} \rightarrow \infty$ for every $q < \infty$ (Phuc [5]; Barker–Seregin [2]), L^3 outside the backward paraboloid blows up, and in the axisymmetric case so does weak- L^3 [1]. If the general weak- L^3 endpoint holds, (H_{out}) fails at every singularity, and every real singularity lies in the *external-supply branch*: unbounded weak- L^3 mass outside the paraboloid, feeding the inner region. That branch, and its dichotomy between energy growth and intense, low-energy spikes, is treated in the companion covering paper [13]. The results here are therefore best read as: *gathering of isolated tubes is excluded wherever the outer field is Type I-like (Theorem 6.2), and for slowly growing local energy whenever Seregin’s weighted bound holds with the natural weight (Proposition 5.1); what remains is an Euler Liouville problem.*

10. OPEN PROBLEMS

Gap 10.1. The Lorentz-space and weight details of Theorem 7.1 are written at the level of a careful sketch; the BFP input has been checked line by line against [1, §4], including the dependence on M through $N(M)$.

Gap 10.2. Proposition 3.1 rests on hypothesis (H) of [11] (coherence of the tube) and on the leak law in its one-sided form; deriving (H) from a profile is part of the covering problem [13].

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ALMATY, KAZAKHSTAN

Email address: adilsalkimbayev@gmail.com